Notes of Database 2

A series of not yet structured notes on the course Database 2 as taught in Politecnico di Milano by Sara Comai during the academic year 2018/2019.

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Concurrency Control

Anomalies

Lost Update $\rightarrow r_1 - r_2 - w_2 - w_1$

Dirty Read $\rightarrow r_1 - w_1 - r_2 - abort_1 - w_2$

Non-Repeat Read $\rightarrow r_1 - r_2 - w_2 - r_1$

Phantom Update $\rightarrow r_1 - r_2 - w_2 - r_1$ (last r_1 reads an inconsistent value)

Phantom Insert $r_1 - w_2(\text{new data}) - r_1$

Core Concepts

- **Goal:**
  reject schedules that cause anomalies
- **Scheduler:**
  a component that accepts or rejects the operations requested by the transaction
- **Serial Schedule:**
  a schedule in which the actions of each transaction occur in a contiguous sequence
- **Serializable Schedule:**
  If it produces the same result of a serial schedule of some transactions.
- **Assumption Commit-Projection:**
  Transaction are observed "in the past" and we decide whether the corresponding schedule is correct.

Class Hierarchy

$$!CSR \implies (\ !2PL \ \wedge \ !TS_MONO)$$

$$CSR \implies VSR$$

$$!2PL \implies !2PL_STRICT$$

VSR - View Serializability

Preliminary Definitions

- <u>*Reads-From*</u>
  $r_i(x)$ ***reads-from*** $w_j(x)$ in a schedule S when $w_j(x)$ precedes $r_i(x)$ and there is no $w_k(x)$ in S between $r_i(x)$ and $w_j(x)$
- <u>*Final Write*</u>
  $w_i(x)$ in a schedule S is a ***final write*** if it is the last write on x the occurs in S

View Equivalence
Two schedules are ***view-equivalent*** $(S_i \approx_{\mathbf{v}} S_j)$ if: they have the same

- *operations*
- *reads-from* relations
- *final writes*

View Serializable
A schedule is ***view-serializable*** if it is view-equivalent to a serial schedule of the same transactions

The class of View-Serializable schedules is named ***VSR***.

Blind Writes

a write $w_i(X)$ is said to be ***blind*** if it is not the last action of resource X and the following action on X is a write $w_j(X)$.

VSR Property each schedule $S \in VSR$ and $S \notin CSR$ has, in its conflict graph, cycles of arcs due to *pairs* of blind writes. These can be swapped without modifying the *reads-from* and *final-writes* relationships.
Once the graph is acyclic it is possible to find a serial schedule view-equivalent to the initial one.

How to find out if a schedule is in VSR?

Do the Conflict Graph.
If such graph is cyclic, consider the cycles that are due to blind writes.
Switch such writes (switch the direction of the arrow).
If now the graph is acyclic we can say it is in VSR.

What's an easy way to build a View Equivalent serial schedule?
Look at the conflict graph and find a node that does not have any incoming arc.
Select it and delete all its outcoming arcs.
Repeat the procedure until you have considered all the nodes of the graph.

CSR - Conflict Serializability

Conflict
Two operations o_i and o_j $(i \neq j)$ are in ***conflict*** if they address the same resource and at least one of them is a write.
Thus, the possible conflicts are:

- *read-write* conflicts
- *write-write* conflicts.

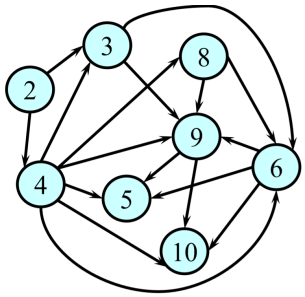
Conflict-Equivalence
Two schedules are ***conflict-equivalent*** $(S_i \approx_{\mathbf{C}} S_j)$ if:
S_i and S_j contain the same operations and all conflicting pairs occur in the same order

Conflict-Serializable
A schedule is ***conflict-serializable*** if it is conflict-equivalent to a serial schedule of the same transactions. The class of conflict-serializable schedules is named ***CSR***.

How to find out if a schedule is in CSR?
We need to do the *Conflict Graph*:

- One node for each transaction T_i.
- One arc from T_i to T_j if there exists at least one conflict between an operation o_i of T_i and an operation o_j of T_j such that o_i precedes o_j.

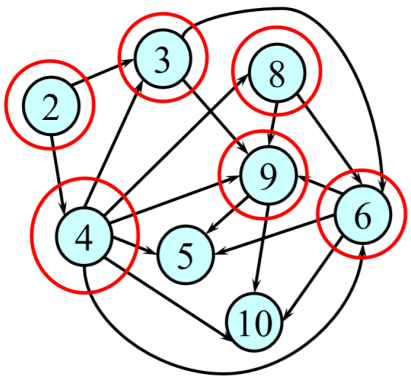
A schedule is in ***CSR*** if and only if its conflict graph is acyclic.



Is there any cycle? No.
The schedule is in CSR (and then also in VSR). The graph suggests, e.g., the serial schedule 2,4,3,8,6,9,5,10 as view equivalent (built choosing and deleting nodes without incoming arcs)

How do we check acyclicity in practice?
A node can be part of a cycle if and only if it has incoming **and** outcoming arcs.
Nodes with only incoming or only outgoing arcs cannot, and can be (recursively) ignored.
The same holds for arc adjacent to such nodes.

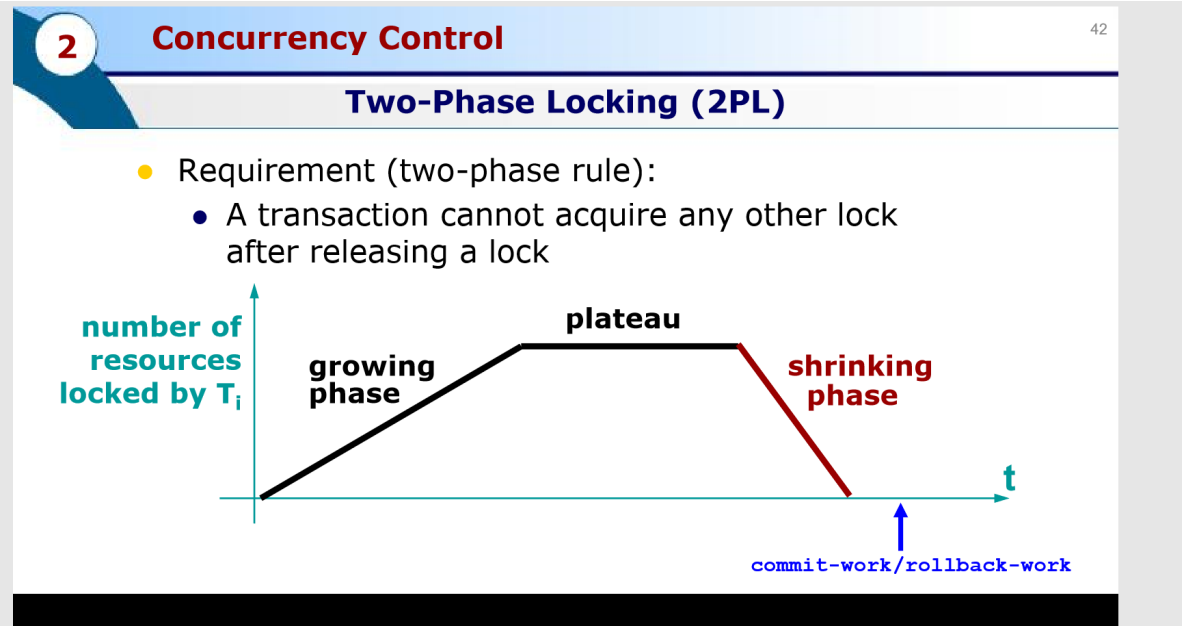
In this way, we not only check for acyclicity, but also identify a serial schedule that is view-equivalent to the given one (if transactions are considered in the order in which the corresponding nodes are deleted)



2PL

OBJECT STATES	PRIMITIVES(OPERATIONS)
free	r-lock
r-locked (1 or more)	w-lock
w-locked (exclusive)	unlock

Rule: a transaction cannot acquire any other lock after releasing a lock



how to actually draw the scheme?
What I've come up with so far is to :

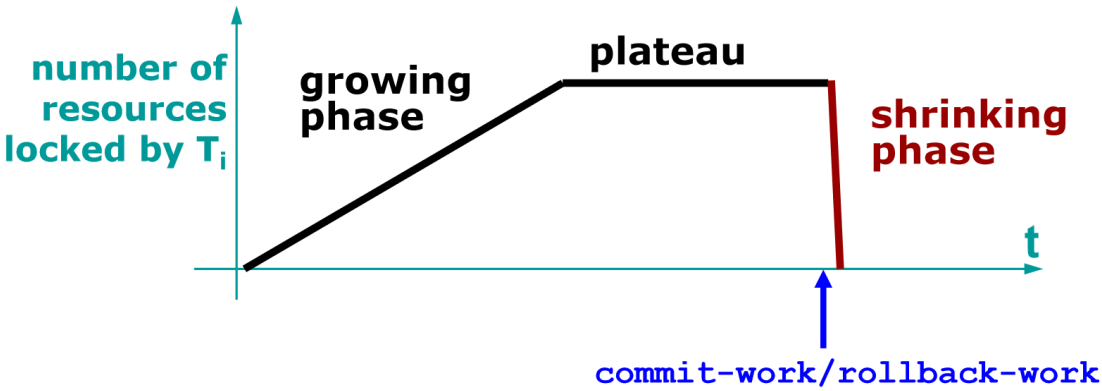
1. list in the page all the read and writes in order of execution (just write them in pencil)
2. under such list create the empty table with transactions as rows.
3. for each row copy the corresponding read and write in the correspond position
4. now complete the table with all the arrows.

2PL Strict

So far we used the hypothesis of *commit-projection* (no transaction in the schedule aborts).
2PL so far does not protect against dirty reads (and therefore neither do VSR nor CSR).

In order to remove such hypothesis we need to add a constraint to 2PL, that defines *2PL Strict*:

*Locks held by a transaction can be released only **after** commit/rollback*.



in 2PL strict, if, for example, I unlock the transition 3, I need to unlock it immediately for all the resources.

$\rightarrow$ in practice it means that in the table with all the read and write I should put all the downward arrows at the end.

TS - Mono

TimeStamp - based concurrency control is characterized by the fact that:

- the scheduler has two counters ***RTM(x)*** and ***WTM(x)*** for each object.
- the scheduler receives read/write requests tagged with timestamps.

Let's clarify:

- $read(x, ts)$:
  - if $ts < WTM(x)$ the request is ***rejected*** and the transaction is killed
  - else, access is ***granted*** and the $RTM(x)$ is set to $\max(RTM(x), ts)$
- $write(x, ts)$:
  - if $ts < RTM(x)$ or $ts < WTM(x)$ the request is ***rejected*** and the transaction is killed
  - else, access is ***granted*** and $WTM(x)$ is set to ts

In order to work without the commit-projection hypothesis, it needs to "buffer" write operations until commit, which introduces waits.

2PL and TS are incomparable, doesn't make sense to compare them since a schedule could be in TS but non in 2PL, the other way around, or even both in TS and in 2PL.

How to find out if a scheduler is in TS in practice?

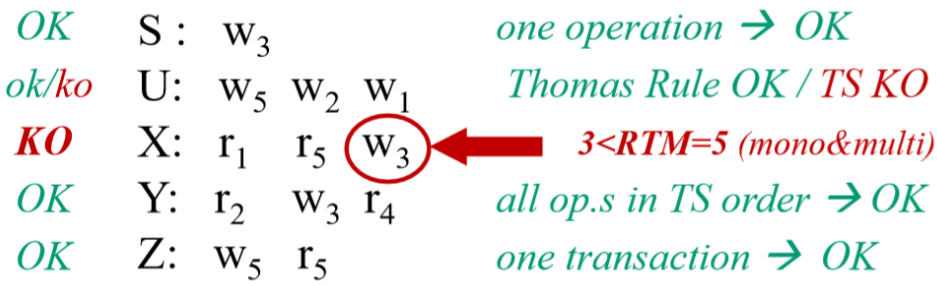
The fastest way consists in considering the transaction resource-wise and do the following.

for each resource we want to have each

- $r - w$
- $w - r$
- $w - w$

pairs ordered increasingly.

Thomas Rule is an improvement to TS which considers legit even resources with such pairs ordered decreasingly.



TS - Multi

- $read(x, ts)$:
  reads are **always accepted**. a copy x_k is selected for reading such that:
  - if $ts > WTM_N(x)$ then $k = N$
  - else take k such that $WTM_k(x) < ts < WTM_{k+1}(x)$
- $write(x, ts)$:
  - if $ts < RTM(x)$ the request is ***rejected*** and the transaction is killed
  - else, a new version is created (N is incremented) with $WTM_N(x) = ts$

How to check TS - Multi in practice?

The fastest way consists in considering the transaction resource-wise and do the following.

if in a row you have r_a ... w_b and $a > b$ then the schedule **is not** in TS-Multi.

XML

XPATH

Basics

Path expression are made of ***steps***. The most common steps are:

- *Nodename*
  - Selects all child nodes of the current node
- */*
  - Selects from the root node
- *//*
  - Selects nodes in the document from the current node that match the selection no matter where they are
- .
  - Selects the current node
- ..
  - Selects the parent of the current node
- @
  - Qualifies attribute names

Conditions

- ***doc("books.xml")/bookstore/book[publisher='Macmillan']/title***
  - returns the title of the books published by Macmillan
- ***doc("books.xml")/bookstore/book[2]***
  - returns the second book of the structure
- ***doc("books.xml")/bookstore/book[2]/***
  - returns all the elements (=*with any tagname) contained into the second book*
- ***[generic node]//@lang***
  - returns all attributes that are named lang
- ***/bookstore/book[1]***
  - selects the first book
- ***/bookstore/book[last()]***
  - selects the last book
- ***/bookstore/book[last()-1]***
  - selects the last but one book
- ***/bookstore/book[position()<3]***
  - selects the first two books
- ***//title[@lang]***
  - selects all the title elements that have an attribute named lang
- ***//title[@lang='eng']***
  - selects all the title elements that have an attribute named lang with a value of 'eng'
- ***/bookstore/book[price>35.00]***

- selects all the book elements of the bookstore element having a price element with a value greater than 35.00

Wildcards

XPath wildcards can be used to select unknown XML elements

- ***
  - matches any element node
- **@***
  - matches any attribute node
- ***node()***
  - matches any node of any kind

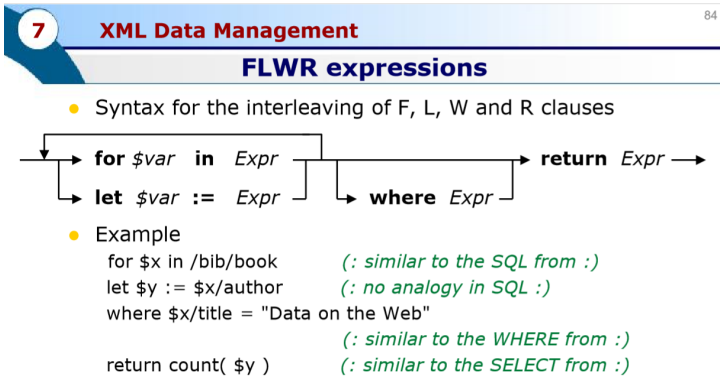
so far example, if I write in a path *//title[@*]*, I will select all title elements which have any attribute

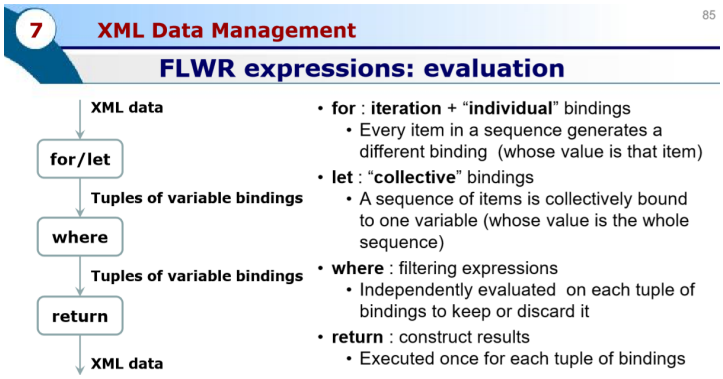
XQuery

FLWR Expressions

FLWR stands for:

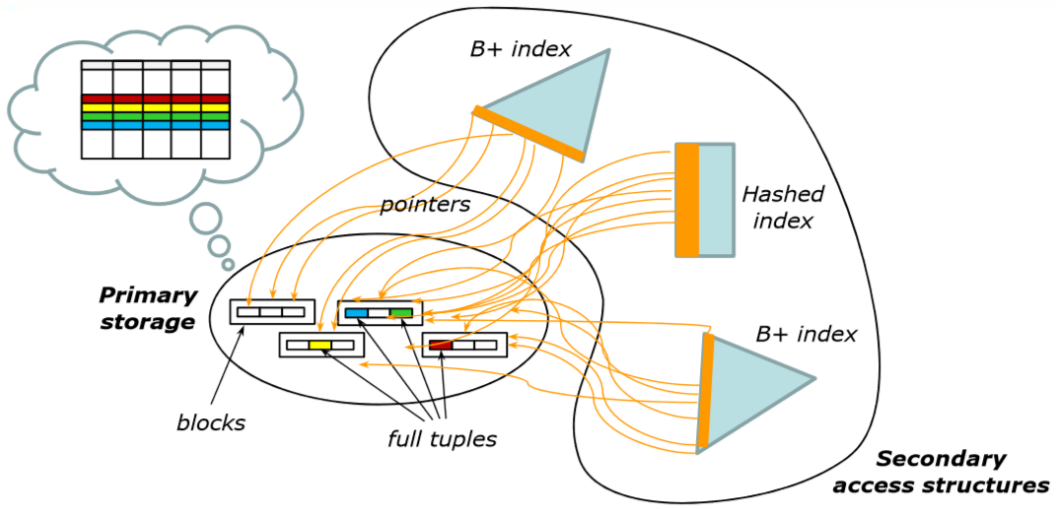
- **F**or
- **L**et
- **W**here
- (Order By)
- **R**eturn





Physical Access Structures Introduction

Each table is stored into exactly one primary physical access structure, and may have one or many optional secondary access structures.



- *Primary Structure*
  - Contains all the tuples of a table
  - Its main purpose is to store the table content
- *Secondary Structures*
  - Used to index primary structures, and only contains the values of some fields, interleaved with pointers to the blocks of the primary structure
  - Their main purpose is to speed up the search for specific tuples within the table, according to some search criterion

	Primary	**Secondary**
Sequential structures	**Typical**	-
Hash-based structures	Frequent	Frequent
Tree-based structures	Frequent	**Typical**

Physical Access Structure Types

Legenda :

- $val(A_j)$ = number of distinct values of the attribute A_j in the table T

Three types :

Sequential Structures

Three types:

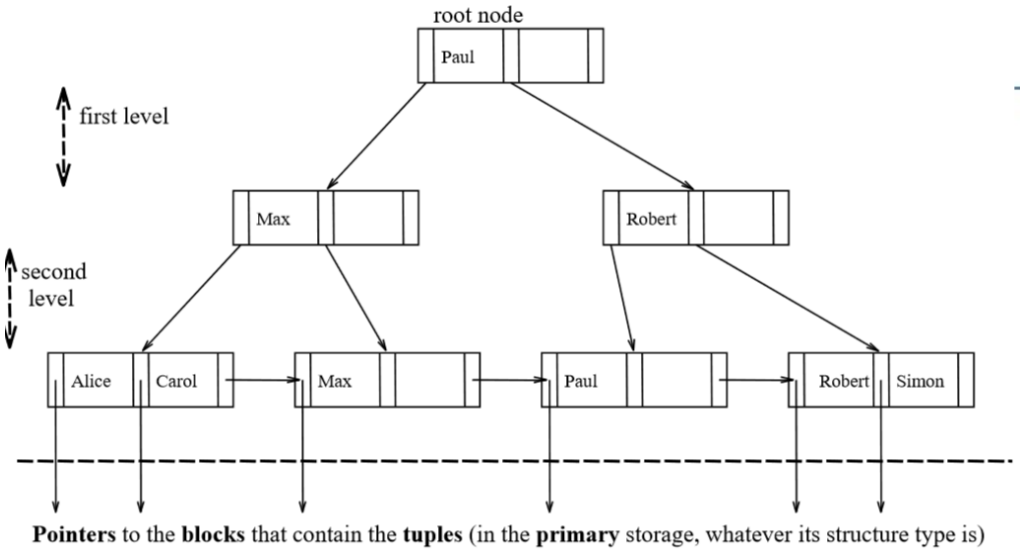
- *Entry - Sequenced*
  - In an Entry - Sequenced organization, the sequence of the tuples is dictated by their order of entry
  - Optimality
    - Optimal for
      1. Space occupancy
      2. Carrying out sequential reading and writing
    - Non - Optimal for
      1. Searching specific data units
      2. Updates that increase the size of tuple, in fact all following tuples must be shifted on.
- *Array*
  - In an array organization, the tuples (all of the same size) are arranged as in an array, and their position depends on the values of an index (or indexes)
  - Appropriate only when the tuples are of fixed length
- *Sequentially - Ordered*
  - In a Sequentially - Ordered organization, the tuples are ordered according o the value of a key (typically one field, but may be obtained by combining more than one attribute)
  - Main problem: Insertions and Updates could increase the data size, this means that such structures require reordering techniques for the tuples already present. There are some techniques to avoid such reordering problem (i.e. leaving a certain number of slots free at the time of first loading, followed by 'local reordering' operations)

Hash - Based Access Structures

Such structures assure an efficient associative access to data, based on the value of a key field to which the hash is applied. Hash - Based structures have N_B buckets, each typically of the size of 1 block. A hash algorithm is applied to the key field so as to compute a value between 0 and $N_B - 1$.

- When is it used? This is the most efficient structure for queries with equality predicates, but it is rather inefficient for interval queries

Tree Structures



Gives associative access based on the value of a key field.

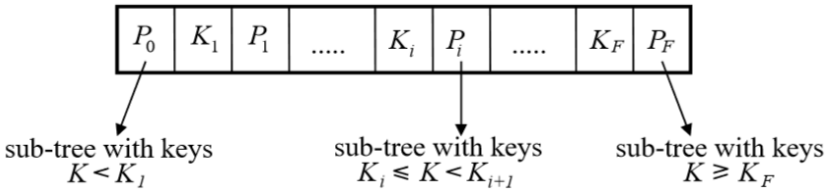
Each tree has:

- One root node
- Several intermediate nodes
- Several leaf nodes

Each node corresponds to a block.

The links between nodes are established by pointers to mass memory.

Search technique :



Suppose we are looking for a tuple with key value V, so at each intermediate node we do the following:

- If $V < K_1$ follow P_0
- if $V \geq K_F$ follow P_F
- Otherwise, follow P_J such that $K_J \leq V < K_j + 1$

The nodes can be organized in two ways:

- *B+ trees*
  - Pointers to the tuples are only contained in the leaf nodes
  - The leaf nodes are linked in a chain ordered by he key
  - Supports interval queries efficiently
  - The most used by relational DBMSs
- *B trees*
  - Pointers to the tuples can also be contained in intermediate nodes. In this way, for tuples with $V = K_i$, we don't kneed to descend the tree down to the leaves.
  - No chain links for leaf nodes, namely there is no support for intervals

Costs

- **Sequential Structures**
  - Cost of Equality Lookups: a full scan, because lookups are no supported
  - Cost of Interval Lookups: a full scan, because lookups are not supported
- **Hash and Tree Structures**
  - Cost of Equality Lookups:
    The lookup is supported only if A_i is the key attribute on which the structure is built, otherwise, a full scan is required. Lookup cost depends on
    - The storage type (primary/secondary)
    - The structure type (tree/hash)
    - The key type (unique/non-unique)

    <u>If it is a primary hash:</u>

    A lookup for $A_i = v$ needs to find the block(s) storing the tuples with the same hash for A_i as those with v. The hash function is applied to v, the right bucket is therefore immediately identified. The bucket contains the searched tuples
    - $Cost = 1$ if the hash has no overflow chains
    - $Cost = 1 + (average\ size\ of\ the\ overflow\ chain)$ if the hash has overflow chains

    <u>If it is a secondary hash:</u>
    The lookup for $A_i = v$ works exactly in the same way, only the aim is to find the block(s) storing the pointers to the tuples with the same hash for A_i as those with v. The Hash function is applied to v, the right bucket is identified, and the corresponding block(s) are retrieved.
    - $Cost =$
      $1(comment:\ for\ the\ initial\ bucket\ block) +$
      $(average\ overflow\ blocks)(comment:\ if\ any) +$
      $(1\ block\ per\ pointer\ to\ be\ followed)$
  - Cost of Interval Lookups:
    lookups based on intervals are not supported
- **B+ Structures**
  - Cost of equality Lookups :
    <u>If it is a primary B+</u>
    A lookup for $A_i = v$ needs to find the block(s) storing all tuples indexed by the key value v. the root is read first, then, a node per intermediate level is read, until the first leaf node that stores tuples with $A_i = v$ is reached. If the searched tuples are all stored in that leaf block, stop. Else, continue in the leaf blocks chain until v changes to v'
    - $Cost =$
      $(1\ block\ per\ intermediate\ level) +$
      $(as\ many\ leaf\ blocks\ as\ necessary\ to\ store\ all\ the\ tuples\ with\ A_i = v)$

    <u>If it is a secondary B+</u>
    The lookup for $A_i = v$ needs to find the block(s) storing all the pointers that point to all tuples with the key value v for A_i. The root is read first, then a node per intermediate level is read, until the first leaf node that sores pointers $A_i = v$ is reached. If the pointers are all stored in that leaf block, stop. Else, continue in the leaf blocks chain until v changes to v'.

- $$Cost = \\ (1\ block\ per\ intermediate\ level) + \\ (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ the\ pointers\ pointing\ to\ the\ tuples\ with\ A_i = v) + \\ (1\ block\ per\ each\ pointer)$$

- Cost of Interval Lookups

  If it is a primary B+
  We consider a lookup for $v_1 \leq A_i \leq v_2$ as the general case. If $(A_i < v)$ or $(v < A_i)$ we just assume that the other edge of the interval is the first/last value in the structure. The root is read first, then a node per intermediate level is read, until the first leaf node that stores tuples with $A_i = v_1$ is reached. if the searched tuples are all stored in that leaf block stop. Else continue in the leaf blocks chain until v_2 is reached.

  - $$Cost = \\ (1\ block\ per\ intermediate\ level) + \\ (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ all\ the\ tuples\ in\ the\ interval)$$

  If it is a secondary B+

  We still consider a lookup for $v_1 \leq A_i \leq v_2$ as the general case. The root is read first, then a node per intermediate level is read, until the first leaf node that stores a the pointers pointing to the tuples with $A_i = v_1$ is reached. If all the pointers (up to v_2) are in that leaf block, stop. Else, continue in the leaf blocks chain until v_2 is reached.

  - $$Cost = \\ (1\ block\ per\ intermediate\ level) + \\ (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ all\ pointers\ to\ the\ tuples\ in\ theinterval)\ + \\ (1\ block\ per\ each\ such\ pointer,\ in\ order\ to\ retrieve\ the\ tuples)$$

Cost of Joins

I cannot talk about the cost of joins regarding each type of structure independently, that's because joins are to do among different tables, that can be represented in different structures one from the other.

Legenda:

- b_A = Number of blocks of the table A
- t_A = Number of tuples of a block of the table A
- E = External table
- I = Internal table
- L = Left
- R = Right

Type of joins:

- *Nested - Loop Join Cost (Assuming not enough free pages to host an entire table in the buffer)*

  compares the tuples of a block of table T_E with all the tuples of all the blocks of T_I before moving to the next block T_E.

  - $Cost = b_E * b_I$
- *Nested - Loop Join Cost (Assuming enough free pages to host an entire table in the buffer)*
  the table in the buffer is chosen as internal table.

  - $Cost = b_E + b_I$

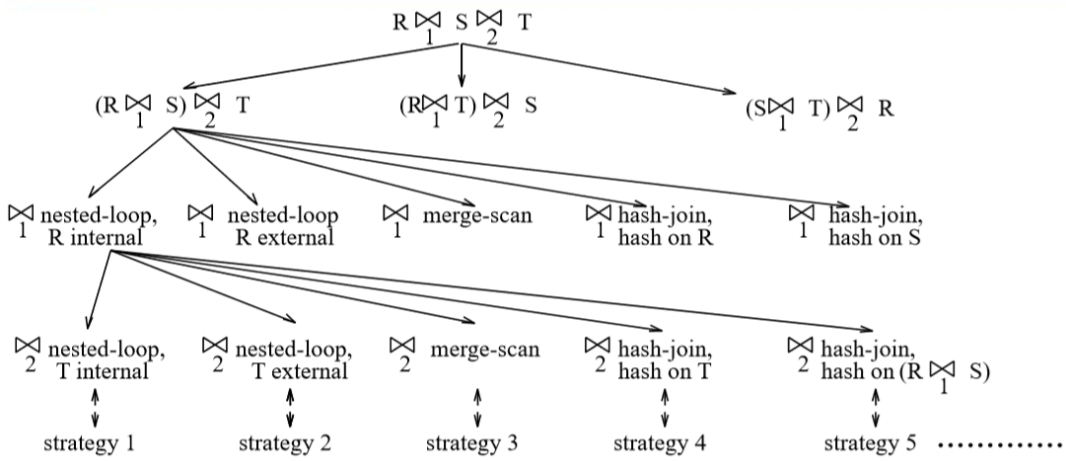
- *Nested - Loop Join Cost (Assuming that one table supports indexed access in the form of a lookup based on the join predicate)*
  such table is chosen as internal
  - $Cost = b_E + (t_E * (cost\ of\ one\ indexed\ access\ to\ T_I)$
- *Merge - Scan Join Cost (Assuming the primary storage to be sequentially ordered or B+)*
  This join is possible only if both tables are ordered according to the same key attribute, that is also the attribute used in the join predicate. The cost is linear in the size of the table.
  - $Cost = b_L + b_R$
- *Hashed Joins (Assuming the two hashes to be both primary storages)*
  This join is possible only if bot tables are hashed according to the same key attribute, that is the same used in the join predicate. The cost is linear in the size of the hashes
  - $Cost = b_L + b_R$

Query Planning

What cost to assign to a plan?

- Construct a decision tree, in which each node corresponds to a choice, each leaf node corresponds to a specific execution plan
- Assign to each plan a cost

$$C_{total} = C_{I/O} n_{I/O} + C_{cpu} n_{cpu}$$

- Choose the plan with the lowest cost, based on operations research (branch and bound)

Example of decision tree: